



REVA Academy for Corporate Excellence

Program: MTech/MS Artificial Intelligence	Course: Deep Learning
Batch: AI 09	Exam: Final
Date: 23 rd Aug 2025	Time: 10:00am to 1:00pm
Duration: 3 Hrs	Marks: 100

Instructions	
1.	This is an open-book exam. However, all the answers must be paraphrased and customized. Any evidence of copying verbatim from any third-party resources or plagiarized content will lead to suspension and other stringent actions. If any identical answers are found, both will be awarded zero.
2.	You may use any tool or software to answer the questions. All the outputs and codebase must be well commented.
3.	All the answers must be written in the answer script template provided by RACE. The template will be available in the LMS with the answer submission link. Along with the main answer script, all other artifacts like screenshots, codebase and any other materials must be submitted. All such artefacts must be named with the corresponding question number.
4.	All the artefacts must be zipped and submitted through the LMS Link. Zipped Folder must be named as 'SRN_Name of the Course'.

Instructions		Answer ONE FULL question from each unit.	Marks	
UNIT - I				
Q1.	a.	Using California Housing Dataset (built into Scikit-learn), demonstrate feature engineering by creating two new features. Provide code and explain why these features could improve model performance.	5	25
	b.	Train two models on the dataset: one using traditional ML (e.g., Logistic Regression or Decision Tree) and another using a simple Deep Neural Network. Compare their accuracy and comment on the role of feature engineering in the results.	10	
	c.	Reflect on a real-world scenario (outside the classroom, such as healthcare, finance, or retail) where deep learning clearly outperforms traditional ML. Support your answer with reasoning and any supporting artefacts (e.g., research/blog reference, code snippet).	10	
OR				
Q2.	a.	Dataset: Use the IMDB Movie Reviews Dataset available in keras.datasets.imdb (50,000 reviews labeled as positive or negative). Load and preprocess the IMDB dataset. Show how reviews are represented as integer sequences. Demonstrate one-	8	25

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		hot encoding of the top 10,000 words and comment on the sparsity of the resulting representation.		
	b.	Using the same dataset, implement word embeddings (via Embedding layer in Keras or pretrained Word2Vec). Compare the dense embeddings with the one-hot encoded representation by visualizing similarities (e.g., most similar words to "good" or "bad").	9	
	c.	Train a simple neural network classifier twice: once with one-hot encoded vectors, and once with embeddings. Report validation accuracy in both cases and critically evaluate why embeddings improve performance for text classification.	8	
UNIT - II				
Q3.	a.	Using the IMDB Movie Reviews Dataset (Keras), preprocess the text and apply Word2Vec or Doc2Vec embeddings to represent reviews numerically. Show how the embeddings capture semantic meaning.	8	25
	b.	Train two sequence models: 1. A simple RNN 2. An LSTM on the IMDB dataset for sentiment classification. Compare their training/validation performance and highlight differences.	9	
	c.	Using Hugging Face’s pretrained model bart-large-mnli, apply zero-shot text classification on 10 review sentences .Compare the predictions with your RNN/LSTM models and critically evaluate the trade-offs of using pretrained LLMs vs training models from scratch.	8	
OR				
Q4.	a.	Use the provided Daily Minimum Temperatures Dataset (Melbourne, Australia). It contains daily minimum temperatures recorded between 1981 and 1990. Load and visualize the dataset. Split it into training and test sets. Perform basic exploratory analysis to identify seasonality or trends in the temperature series.	5	25
	b.	Build and train a basic/plain LSTM model to predict the next day’s temperature. Show the training and validation loss curves. Report model performance using RMSE or MAE.	10	
	c.	Experiment with at least one improvement technique (e.g., adding more LSTM layers, using Dropout for regularization, or tuning window size). Compare the new model’s results with the baseline LSTM and critically evaluate the difference.	10	
UNIT - III				

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Q5.	a.	Dataset Kaggle: Sports Balls – Multiclass Image Classification Description: ~9,000 images of sports balls across 15 classes, including basketball, cricket ball, tennis ball, rugby ball, etc. Build and train a basic CNN from scratch (e.g., 3–4 convolutional + pooling layers) to classify images into 15 sports-ball categories. Report training and validation accuracy and show at least one confusion matrix for analysis.	5	25
	b.	Incorporate data augmentation (e.g., rotations, flips, zooms) during training. Re-train the model and compare performance (accuracy, loss curves) with the non-augmented version.	10	
	c.	Critically evaluate why transfer learning + fine-tuning often outperforms training from scratch, especially when datasets have many target classes but limited images per class. Support your answer with references or case study examples.	10	
	OR			
Q6.	a.	Dataset: Use the CIFAR-100 dataset available in keras.datasets.cifar100. (a) Build and train a basic CNN from scratch (3–4 convolutional + pooling layers) to classify CIFAR-100 images. Report training and validation accuracy.	5	25
	b.	Apply data augmentation (rotation, flipping, cropping, etc.) and re-train the same CNN. Compare results with the non-augmented model.	10	
	c.	Critically discuss the challenges of training CNNs on a dataset with a large number of classes (100). Suggest at least two strategies (e.g., deeper architectures, transfer learning, label hierarchy) to address these issues.	10	
	UNIT - IV			
Q7.	a.	Dataset :Agriculture_FAQ_dataset.csv Build a basic Retrieval-Augmented Generation (RAG) chatbot that takes user queries (e.g., “What is the best time to plant rice?”) and retrieves relevant answer from the dataset, then provides answer using a large language model (Mistral, Mixtral or any one of your choice).	10	25
	b.	Implement and compare at least two retrieval methods (e.g., TF-IDF vs. Sentence Transformers embeddings). Show which method gives more relevant responses with examples.	10	
	c.	Critically reflect on two real-world challenges in deploying such a RAG chatbot (e.g., hallucination, outdated information, high inference cost) and propose possible mitigations.	5	
	OR			
Q8.	a.	Dataset Description: A collection of ~1,500 wildlife images featuring four animal classes — buffalo, elephant, rhino, and zebra.	10	25

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		https://www.kaggle.com/datasets/ankanghosh651/object-detection-wildlife-dataset-yolo-format Load the Wildlife dataset and train a YOLOv8 model to detect the four animal classes. Display detection results on at least 5 test images.		
	b.	Track key metrics during training — specifically mAP@0.5 and mAP@0.5:0.95 — and interpret what they indicate about the model's detection performance.	5	
	c.	Propose at least two enhancements (e.g., data augmentation strategies, tuning architecture hyperparameters, or fine-tuning pre-trained weights) that could significantly improve detection accuracy. Justify your choices with reasoning or code examples.	10	
